AI-Assisted Development — Exam Submission

HotFixAmbulance — AI-Powered Production-Error Triage for .NET Web APIs

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AI tools used: Claude Code (primary)

Repository: https://github.com/myPOStech/mps-banking-hot-fix-ambulance

Working-system evidence: `docs/evidence/working-system-evidence.pdf`

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1. Project Overview

The idea

On-call engineers waste the first, most expensive minutes of an incident doing the same manual ritual: open Kibana, filter the last few hours of errors, eyeball the stack traces, guess the root cause, and go hunting through Git for the offending change. **HotFixAmbulance** automates that ritual. It is a triage tool that, for a chosen .NET Web API and time window, pulls the recent error logs, groups them into distinct problems, and — for each problem — uses a **local Qwen LLM** to write two human-readable columns: **"Suggestion for Error"** (what the error means) and **"How to fix"** (a concrete remediation), grounded in the service's own Git history. The result is a ranked, explained, actionable error table in a web UI, plus a CLI for terminal/CI use.

System requirements

Functional

- Fetch Fatal/Error/Warning logs for a named API over a relative (last *N* h) or absolute time window from **Elasticsearch**.
- Group logs by exception fingerprint (exception type + normalized message + endpoint) and rank by severity, count, recency.
- For each group, produce an AI **Suggestion** and **How-to-fix**, using a **locally hosted Qwen model** and grounding the answer in Git blame + related commits.
- Persist every run; expose it through a REST API and a paginated React table; provide a CLI that triages and prints JSON/table output.
- The UI must make it **visible and verifiable** that a group was analysed by the LLM (a per-group 🤖 **Qwen** badge).

Non-functional

- **Graceful degradation:** the LLM is optional — if the model is unreachable, the system silently falls back to a deterministic Git-history heuristic and never fails a run.
- **Local & private:** the model runs on the developer's machine in Docker (CPU-only), no data leaves the host.
- **Reproducible:** one command (`scripts/demo.ps1`) brings up the whole stack (Qwen, Elasticsearch, SQL Server), generates sample errors, runs triage, and opens the UI.

Tech stack

.NET 9 / C# 13 (backend, CLI) · ASP.NET Core Minimal APIs · Entity Framework Core + SQLite · Elasticsearch 8 · LibGit2Sharp · React 18 + TypeScript + Vite + TanStack Table + Tailwind · Ollama-runtime serving **Qwen 2.5 (3B)** · Docker Compose · PowerShell automation · xUnit + FluentAssertions + NSubstitute (backend) · Vitest + Testing Library (frontend).

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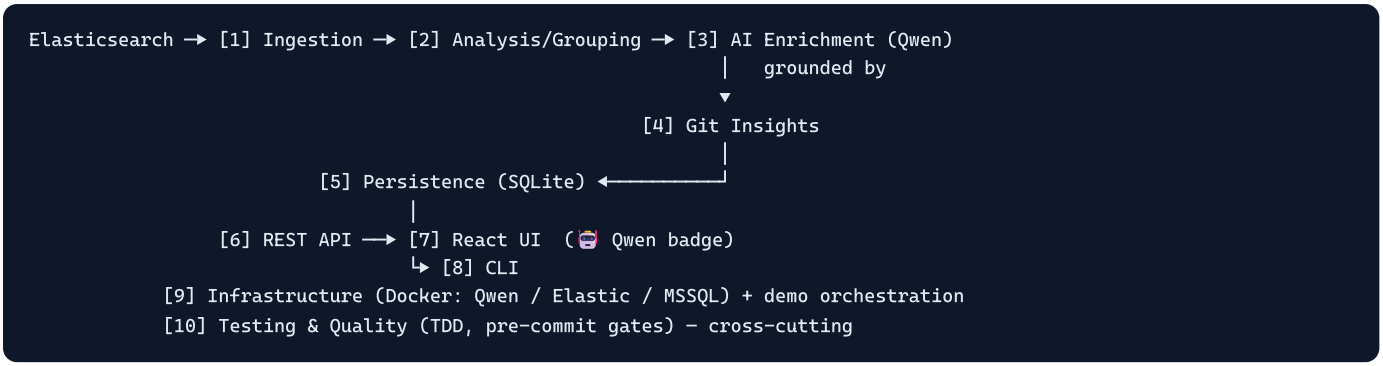
2. System Architecture — Modules

The system was decomposed (with AI assistance) into focused technological modules, each a separate .NET project or front-end/infra unit with a single responsibility and a clear seam (interface) to its neighbours.



#	Module	Project / Location	Responsibility
1	Log Ingestion	`HotFixAmbulance.Elastic`	Query Elasticsearch, normalise documents to `LogEntry`.
2	Analysis & Grouping	`HotFixAmbulance.Analysis`	Group by fingerprint, rank, baseline suggestions.
3	AI Enrichment (Qwen)	`HotFixAmbulance.Llm` + `LlmGroupEnricher`	Per-group LLM call; JSON contract; fallback.
4	Git Insights	`HotFixAmbulance.GitInsights`	Blame + related commits to ground the model.
5	Persistence	`HotFixAmbulance.Persistence`	Store runs + per-group `AnalyzedBy` in SQLite.
6	REST API	`HotFixAmbulance.Api`	Endpoints for triage, history, paged groups.
7	Frontend UI	`frontend/`	Paginated table; renders the Qwen badge.
8	CLI	`HotFixAmbulance.Cli`	One-shot triage for terminal / CI.
9	Infrastructure	`infra/`, `scripts/`	Dockerised Qwen/Elastic/MSSQL + demo runner.
10	Testing & Quality	`tests/`, `.claude/hooks`	TDD, unit/integration tests, pre-commit gates.

3. Development Process per Module

> *For every module:* ***Approach & reasoning***, ***Step-by-step workflow***, ***Testing strategy***, ***AI tool choice***, *and representative prompts.*

3.1 Log Ingestion (Elasticsearch)

Approach & reasoning. Keep Elasticsearch behind an `IElasticLogSource` seam so the pipeline depends on a domain `LogEntry`, not on the ES client. A `SerilogDocumentMapper` translates the stored document shape into `LogEntry`; an `ElasticLogIngestor` validates inputs, applies the severity filter and a `MaxDocuments` cap.

Step-by-step workflow. Asked the AI to (1) define `LogEntry` / `TimeWindow` / `LogQuery` value types, (2) write the ingestor with relative/absolute window support and an `IsTruncated` flag, (3) write the ES client adapter.

Testing strategy. Unit tests with a substituted `IElasticLogSource` cover the window math, severity filter and truncation; an Elasticsearch ingest pipeline (`infra/elasticsearch`) is bootstrapped for the live demo.

AI tool choice. Claude Code — it could hold the whole ingestion contract in context and generate the mapper + tests together.

Prompts: *"Design a LogEntry domain type and an ingestor that filters Fatal/Error/Warning over a UTC window with a max-documents cap."* · *"Write unit tests for the truncation flag using a fake log source."*

3.2 Error Analysis & Grouping

Approach & reasoning. Grouping must be **deterministic** (no LLM), so it lives behind `IAnalysisStrategy`. `HeuristicAnalyzer` groups by *(exception type, normalized message, endpoint)*, ranks by severity → count → last-seen, and fills baseline AI columns from rule matches.

Step-by-step workflow. (1) `MessageNormalizer` to collapse volatile IDs; (2) `ErrorGroup.FromLogs` aggregation; (3) `RankBySeverity`; (4) a `SuggestionBuilder` + `DefaultRules` baseline.

Testing strategy. Pure unit tests assert grouping keys, ranking order and correlation-id counting — fast and exhaustive because the layer is side-effect-free.

AI tool choice. Claude Code — strong at generating table-driven ranking tests and edge cases.

Prompts: *"Group logs by a stable fingerprint and rank Fatal>Error>Warning, then count desc, then last-seen desc."* · *"Add a normalizer that replaces GUIDs/numbers so the same error collapses into one group."*

3.3 AI Enrichment with Qwen (the centrepiece)

Approach & reasoning. The AI columns are owned by an `IGroupEnricher` seam so the strategy is swappable from config (`Analysis:Strategy`). `LlmGroupEnricher` builds a strict-JSON prompt (`{suggestion, howToFix}`) from the group + Git evidence, calls the model via `ILlmClient` (`OllamaLlmClient` , `temperature 0.2` , `format:"json"`), and on **any** failure (timeout, unreachable, bad JSON) degrades to the deterministic Git heuristic. A successful call tags the group `AnalyzedBy = "Llm"` ; that single fact drives the UI badge.

Step-by-step workflow. (1) `LlmPromptBuilder` (system contract + fact dump); (2) `OllamaLlmClient` that never throws (returns `null` on failure); (3) `LlmGroupEnricher` with fallback; (4) propagate per-group `AnalyzedBy` through `TriageService` → `ErrorGroup` → API → UI.

Testing strategy. Unit tests with a substituted `ILlmClient` assert: valid JSON → `Llm` ; null/garbage → heuristic fallback; and that `TriageService` stamps each group with the enricher's source.

AI tool choice. Claude Code — it implemented the fallback contract, the run-level `Mixed/Llm/Heuristic` aggregation, and the tests in one TDD loop.

Prompts: *"Add an LLM enricher that asks the model for {suggestion, howToFix} as JSON and falls back to the Git heuristic on any error — never throw."* · *"Make sure every group records which strategy produced it, and surface a per-group marker in the UI proving Qwen was used."*

3.4 Git-History Grounding

Approach & reasoning. An LLM answer is only useful if grounded in the real code, so `FixHintBuilder` (behind `IFixHintSource`) uses LibGit2Sharp to fetch the offending file's blame + a code snippet and search `origin/main` for related commits; this evidence is fed to both the heuristic and the Qwen prompt.

Step-by-step workflow. (1) `IGitRepoCache` to clone/pull per API; (2) `IGitHistoryReader` for blame + keyword commit search; (3) `FixHintBuilder` to assemble a compact evidence string.

Testing strategy. Unit tests substitute the repo cache/reader and assert the formatted hint contains the expected `file:line` , blame and commit SHAs.

AI tool choice. Claude Code — handled the LibGit2Sharp specifics and kept the evidence format testable.

Prompts: *"Given a stack file+line, return blame + a code snippet from origin/main and the related commits."*

3.5 Persistence & REST API

Approach & reasoning. Runs are stored as a row with the groups serialised to JSON (SQLite), keeping the schema trivial and migration-free; the API is ASP.NET Core Minimal APIs with server-side paging/sorting (`GroupPager`).

Step-by-step workflow. (1) `HotFixDbContext` + `TriageRun` (incl. nullable `AnalyzedBy` for back-compat); (2) repository; (3) endpoints: run triage, latest, by-id, paged groups, history.

Testing strategy. Integration tests with `WebApplicationFactory` hit the real endpoints (in-memory/SQLite), asserting status codes, paging bounds and the `analyzedBy` field round-trips.

AI tool choice. Claude Code — generated endpoints + integration tests and kept DTOs in sync with the frontend types.

Prompts: *"Expose paginated groups with sort/dir validation and an allowed page-size list, plus a run-by-id endpoint."*

3.6 Frontend UI + the Qwen badge

Approach & reasoning. A TanStack-Table grid with toggleable columns; the AI columns render a small violet 🤖 **Qwen** badge **iff** `analyzedBy === 'Llm'` — the visible proof the model was used.

Step-by-step workflow. (1) typed API client + `ErrorGroup` type incl. `analyzedBy`; (2) `QwenBadge` component; (3) render it in the Suggestion and How-to-fix cells; (4) metrics panel ("AI Insights Generated").

Testing strategy. Vitest + Testing Library: badge appears for `Llm`, is absent for `Heuristic` / `null` — written **before** the component (TDD red→green).

AI tool choice. Claude Code — wrote the failing tests first, then the component, matching the repo's TDD rule.

Prompts: *"Render a Qwen badge on the AI columns only when analyzedBy is 'Llm'; add Vitest tests for all three cases first."*

3.7 Infrastructure & DevOps

Approach & reasoning. Mirror the existing `infra/<service>` pattern: a CPU-only Qwen runtime in Docker Compose (`infra/qwen`), an idempotent `bootstrap.ps1` that pre-pulls `qwen2.5:3b`, and a `demo.ps1` that orchestrates the whole stack and **asserts** the produced run was analysed by the LLM before opening the UI.

Step-by-step workflow. (1) compose + healthcheck; (2) bootstrap (pull + chat probe); (3) corporate-CA injection (`export-host-ca.ps1` + `docker-compose.corp.yml`) for TLS-intercepting proxies; (4) wire `-WithLlm`/default LLM env into `demo.ps1`.

Testing strategy. Each script is verified live: compose `config`, container health, `/api/tags`, model present, idempotent re-run, and an end-to-end demo asserting `analyzedBy ∈ {Llm, Mixed}`.

AI tool choice. Claude Code — it ran the Docker/PowerShell commands, diagnosed failures from their output, and iterated.

Prompts: *"Make Qwen part of the infrastructure in Docker like Elasticsearch, and guarantee the model is present before the app starts."* · *"The container can't verify TLS through our proxy — inject the corporate CA."*

3.8 Testing & Quality (cross-cutting)

Approach & reasoning. A mandatory TDD cycle (failing test first) and a **pre-commit hook** that blocks any commit unless `dotnet test`, `dotnet format`, frontend tests and lint all pass — so `main`/integration stays green.

Step-by-step workflow. Write the failing test → minimal implementation → run → commit (hook gates it).

Testing strategy. 160 backend unit tests + 18 integration tests + 34 frontend tests, all green at each commit.

AI tool choice. Claude Code — drove the red→green→commit loop and respected the project's skills/hooks.

Prompts: *"Add the failing test for AnalyzedBy first, watch it fail, then implement and run the suite."*

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4. Challenges & Tool Comparison

Biggest challenges

- **Corporate TLS interception.** The Docker container could not `ollama pull` the model — `x509: certificate signed by unknown authority` — because the corporate proxy MITM-terminates TLS and the container lacked the corporate root CA. **Solved** by exporting the host CA bundle (`export-host-ca.ps1`) and mounting it via a `docker-compose.corp.yml` overlay with `SSL_CERT_FILE`. This was the single hardest, most environment-specific problem.
- **A latent DI bug surfaced by the new feature.** The CLI had never been wired to the `IGroupEnricher` seam added in a prior phase, so it crashed with *"Unable to resolve service for type IGroupEnricher."* Found it by running the CLI; fixed by registering `AddHotFixGroupEnrichment` — which also enabled the CLI to use Qwen.
- **CPU inference latency.** One LLM call per group, sequential, on CPU (~10 s each) makes a 38-group run take minutes; mitigated by warming the model and raising the API client timeout. A real trade-off of running a private model locally.
- **Tooling papercuts.** PowerShell 5.1 mis-parsing a UTF-8 script because of em-dash characters (replaced with ASCII); Chrome headless needing absolute paths and an origin before `localStorage`; SQLite file-locks when a running API blocked the build. Each was diagnosed from command output and fixed.

Which tool helped the most, and why

Claude Code was the primary and most valuable tool. Beyond code generation it could **run** the project — execute

Docker/PowerShell/dotnet, read the failures, and iterate — which is what actually solved the infrastructure problems (the CA injection, the DI crash, the parse bug). Its skills/sub-agents kept the work disciplined (plan → TDD → verify → commit with a passing pre-commit gate), and it could reason across the whole repo (backend + frontend + infra) to keep types, DTOs and the demo orchestration consistent end-to-end.

What I would improve next

- **Batch / parallelise** the per-group LLM calls (or stream) to cut run time; optionally cache by fingerprint.
- Add a **GPU compose overlay** for much faster inference where a GPU is available.
- Make the persisted strategy tag itself read `Qwen` (currently the internal tag stays the generic `Llm`, while the UI shows "Qwen").
- Share one SQLite store between CLI and API so a CLI-produced run is viewable in the UI without a second pass.
- Add a small **eval harness** that scores the model's suggestions against known root causes.

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5. Working System Evidence

See `docs/evidence/working-system-evidence.pdf` (4 captioned figures). Highlights:

- **Figure 1 — Overview:** a real run for `demo-api` — **73 logs → 38 groups**, with **38 "AI Insights Generated"** and **38 "Fix Recommendations."**
- **Figure 2 — Proof of Qwen:** every AI cell shows the 🤖 **Qwen** badge and genuine, context-specific model text

(e.g. *"…failed database save operation at line 311 in DemoDatabase.cs…"*, *"…invalid SQL object name 'Pricing.Records'…"*).

- **Figure 3 — Full table:** all **38 groups** carry the badge (run-level `analyzedBy = "Llm"`).
- **Figure 4 — Terminal:** `docker ps` (Qwen healthy on :11434), `ollama list` (`qwen2.5:3b`, 1.9 GB), API run `analyzedBy = Llm` with `withSuggestions = 38` / `withFixes = 38`, and the container at **1177 % CPU** during inference.

Reproduce locally:

```
powershell -File scripts/demo.ps1 -WithElastic -KeepRunning
# then open the printed UI URL — every analysed group shows the 🤖 Qwen badge
```

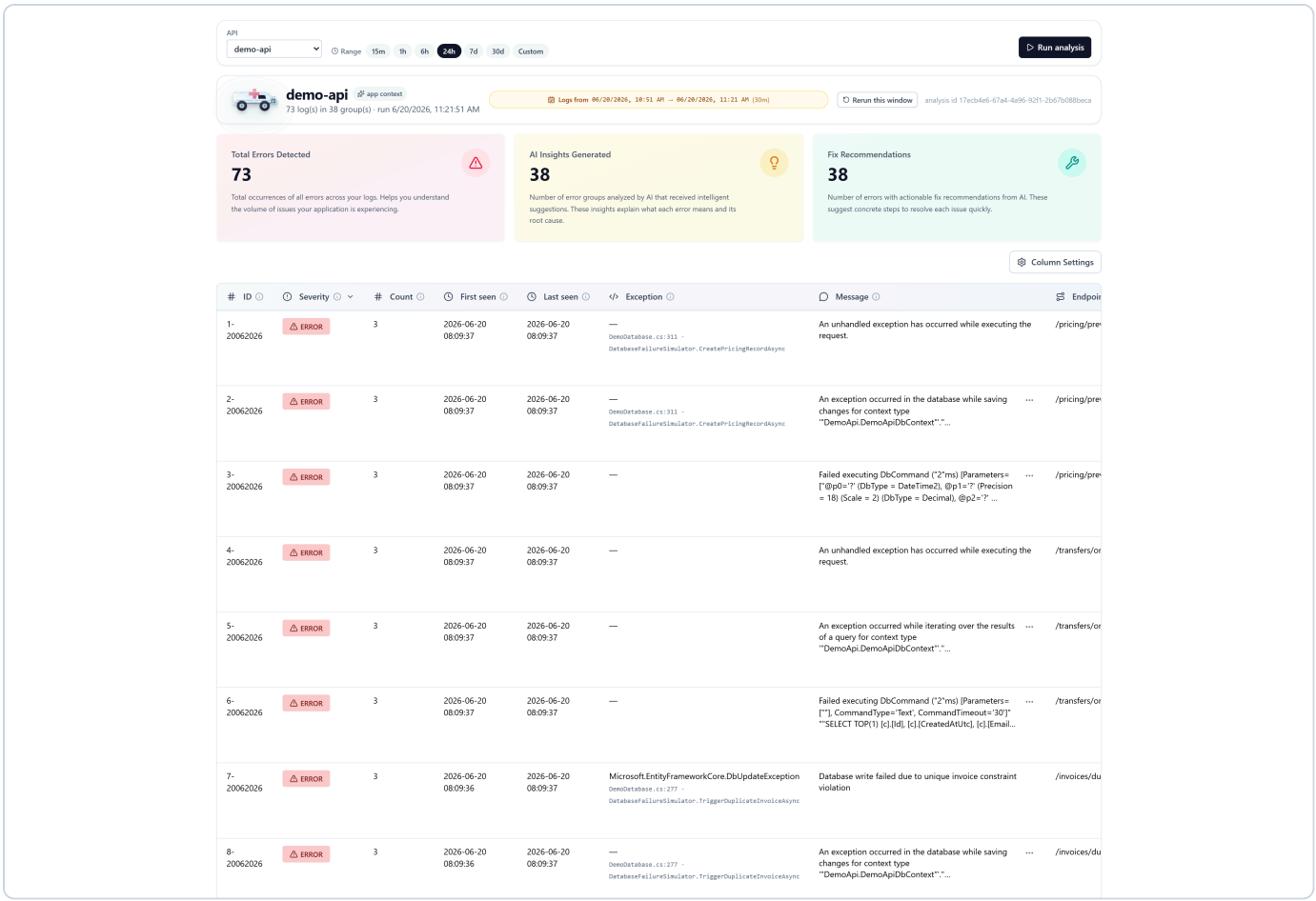

Figure 1 — System overview (ingestion, grouping, AI layer ran). A real triage run for the **demo-api** service: **73 error logs** collapsed into **38 distinct groups**. The metrics panel confirms the AI layer ran end-to-end — **38 "AI Insights Generated"** and **38 "Fix Recommendations"** (every group enriched). The table shows each group's severity, count, first/last-seen, exception with stack frame, message and endpoint — the raw evidence read from Elasticsearch. *Proves the ingestion → grouping → analysis pipeline works.*

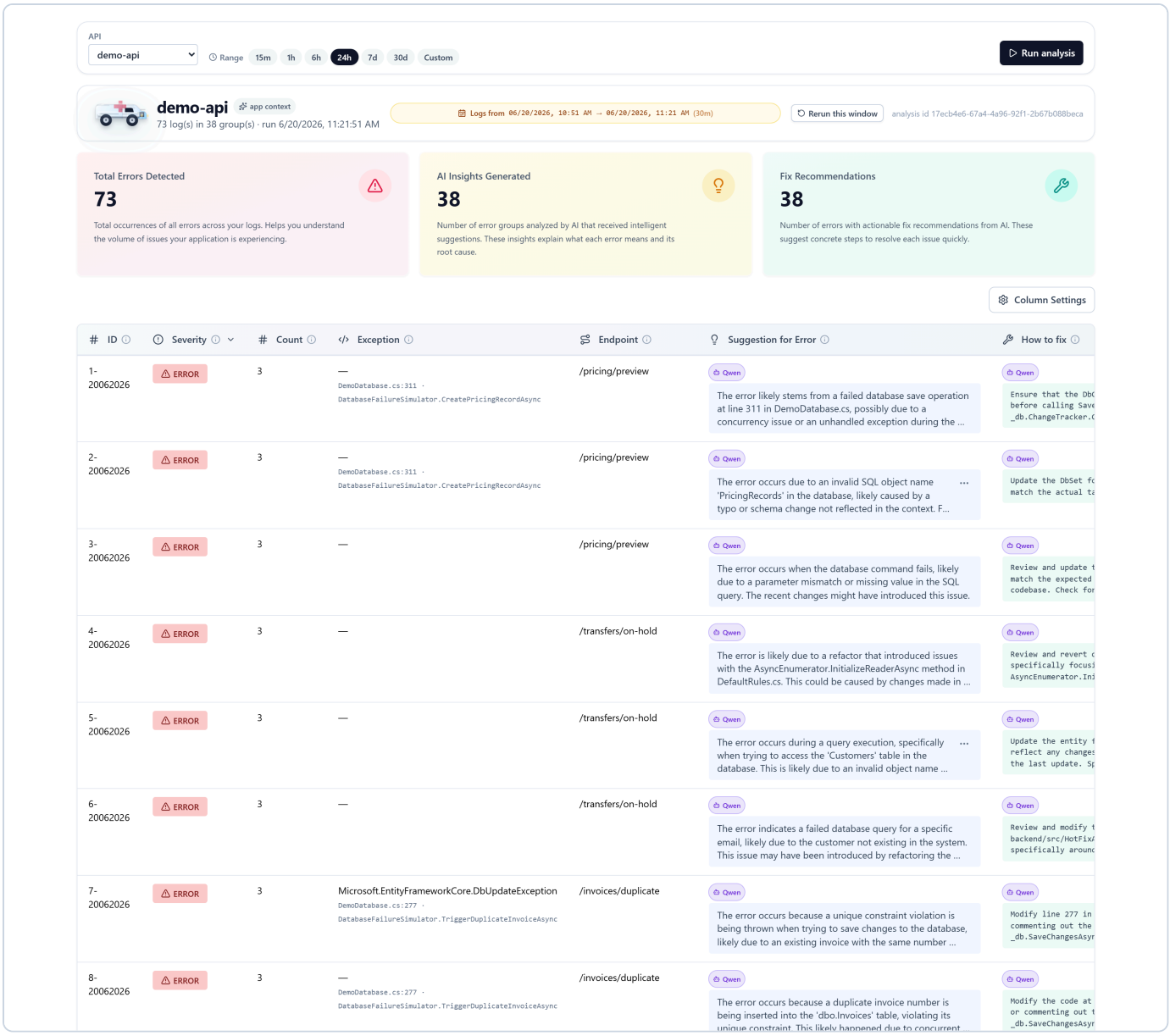

Figure 2 — Proof the analysis is produced by Qwen. The two right-most columns are the AI columns; each cell carries a violet Qwen badge, rendered *only* when a group's `analyzedBy === 'Llm'` — a value the backend sets exclusively after a successful Qwen completion. The text is genuinely model-generated and context-specific (e.g. "...failed database save operation at line 311 in DemoDatabase.cs...", "...invalid SQL object name 'Pricing.Records'..."), not a static template. *Proves the system demonstrably uses Qwen.*

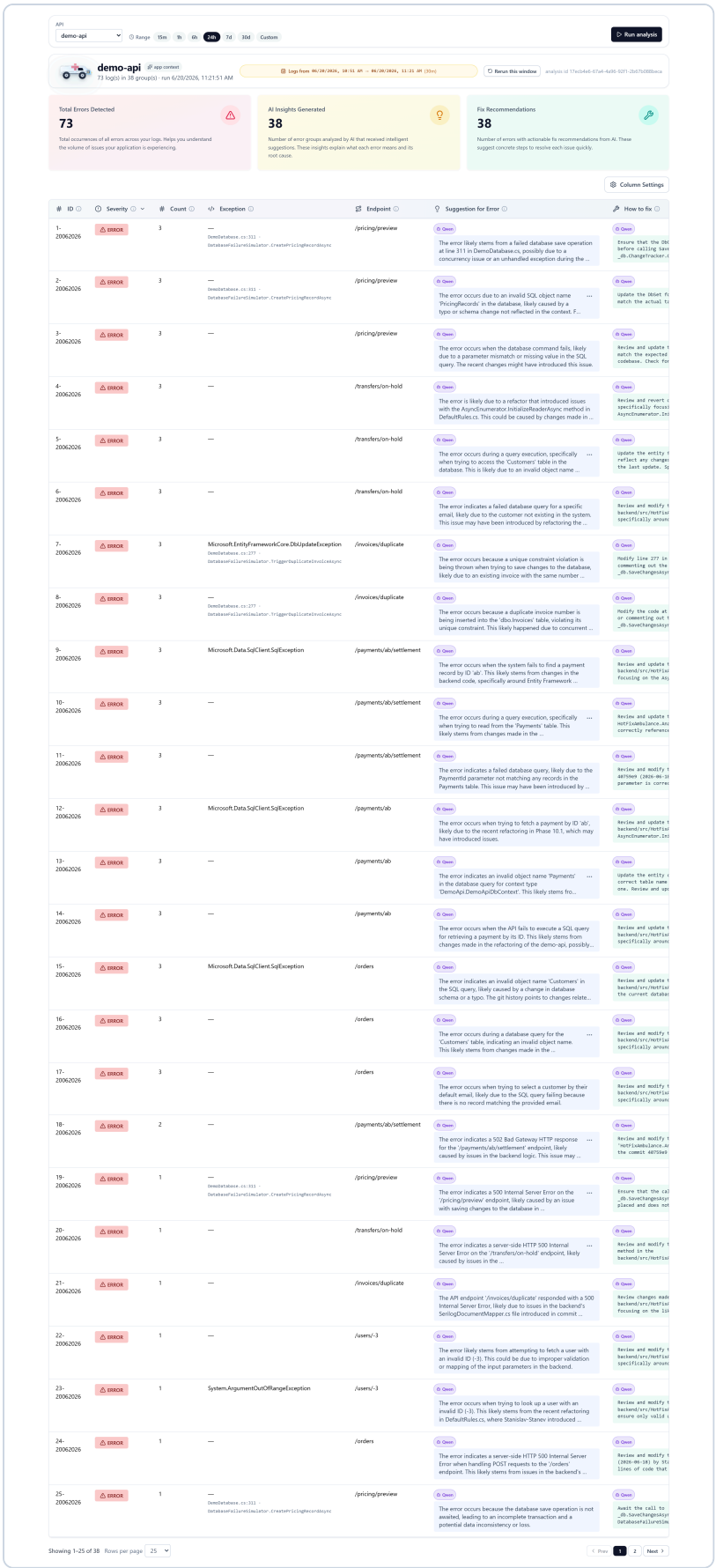

Figure 3 — The complete result set. All **38 groups** carry the Qwen badge on both AI columns (run-level `analyzedBy = "Llm"`), confirming the whole run was analysed by Qwen, not the heuristic fallback.

Figure 4 — Backend / runtime evidence (terminal).

```
# Qwen is served from Docker (CPU-only) on :11434
$ docker ps --filter name=hfa-qwen
hfa-qwen   ollama/ollama:latest   Up 59 minutes (healthy)   0.0.0.0:11434->11434/tcp

# qwen2.5:3b is pulled into the persistent volume
$ docker exec hfa-qwen ollama list
NAME          ID              SIZE     MODIFIED
qwen2.5:3b    357c53fb659c    1.9 GB   37 minutes ago

# A triage run produced via the API — analysed by the LLM, every group enriched
$ curl -s http://localhost:5283/api/triage/runs/17ecb4e6-.../
apiName     = demo-api
analyzedBy  = Llm
totalLogs   = 73
totalGroups = 38
summary.withSuggestions = 38      # all 38 groups got an AI suggestion
summary.withFixes       = 38      # all 38 groups got an AI fix

# During inference the Qwen container saturates the CPU (one call per group)
$ docker stats hfa-qwen --no-stream
hfa-qwen   CPU = 1177%   MEM = 4.4 GiB / 15.4 GiB
```

Proves Qwen runs locally in Docker, the model is loaded, and a real run returns `analyzedBy = Llm` with an AI suggestion + fix for every one of the 38 groups.

6. Repository

GitHub: https://github.com/myPOStech/mps-banking-hot-fix-ambulance

(Feature work on branch `integration-llm`.)

7. Author's Notes

This is **my own idea project**. Beyond the exam, I intend to present it to **myPOS management** as a **working prototype** for everyday use by the **.NET teams** — a practical, privacy-respecting way to shorten incident triage in a fintech environment where production data must stay in-house.

Why it is worth adopting:

- **Faster incident response (lower MTTR).** On-call engineers open a ranked, *explained* error list instead of scrolling raw Kibana logs — the "what" and the "how to fix" are already written for each distinct problem.
- **Lower skill barrier.** Junior engineers get plain-English root-cause hints and remediation steps **grounded in the team's own Git history** (blame + related commits), not generic advice.
- **Consistent, repeatable triage.** Every incident is analysed the same way, with the same evidence, so handovers and post-mortems start from a common baseline.
- **Privacy & compliance by design.** The model runs **locally in Docker (CPU-only)** — no logs, stack traces or source context leave myPOS infrastructure. Critical for banking/PCI contexts where cloud LLMs are a non-starter.
- **No per-token cost.** Uses a small open model (`qwen2.5:3b`) locally; there are no cloud-API usage fees.
- **Fits the existing stack.** .NET, Serilog, Elasticsearch and Git are tools the teams already run; adoption is additive, not a migration.
- **Safe to roll out.** The LLM is optional and **degrades gracefully** — if the model is down, triage still works via the deterministic Git heuristic, so it can be enabled per-team without risk.
- **Extensible.** Pluggable analysis strategy (heuristic ↔ LLM), pluggable model and per-API configuration mean it can grow from a prototype to a shared internal service.

Proposed next step at myPOS: pilot it with one .NET team on a non-critical service for two weeks, measure the

change in triage time and engineer feedback, then decide on a shared, GPU-backed internal deployment.

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8. How to Reproduce the Full End-to-End Test

This section lets a reviewer reproduce the **entire experimental setup** shown in §5 from a clean checkout. One PowerShell command brings up the whole stack (Qwen + Elasticsearch + SQL Server), generates sample errors, runs the triage on Qwen, starts the API + UI, asserts the run was analysed by the LLM, and opens the browser on it.

8.1 Prerequisites

- **Windows 10/11 + PowerShell 5.1** (the automation uses Windows/PowerShell features).
- **Docker Desktop** installed and **running** (Linux containers).
- **.NET 10 SDK**, **Node.js 18+ & npm**, **Git**.
- **Disk/network:** the first run downloads ~**6 GB** of images + models (Ollama image, `qwen2.5:3b` ≈ 2 GB, Elasticsearch, SQL Server). Subsequent runs reuse the cached volumes and are much faster.
- A free `11434` (Qwen), `5283` (API), `5173` (frontend), `9200` (Elasticsearch), `14333` (SQL Server).

8.2 One-time setup

```
# 1) Clone and switch to the feature branch
git clone https://github.com/myPOStech/mps-banking-hot-fix-ambulance.git
cd mps-banking-hot-fix-ambulance
git checkout integration-llm

# 2) Restore tooling and install the pre-commit hook (.NET restore + npm install)
powershell -File scripts/bootstrap.ps1

# 3) Make sure Docker Desktop is running
docker version
```

8.3 Run the full end-to-end demo

```
powershell -File scripts/demo.ps1 -WithElastic -KeepRunning
```

This single command performs, in order:

1. **Starts the Qwen runtime** (`infra/qwen` , Docker, CPU-only) and runs its bootstrap, which **pulls** `qwen2.5:3b` if absent and verifies it with a JSON `/api/chat` probe. *(First run downloads ~2 GB — be patient.)*
2. Sets `Analysis:Strategy=Llm` + the `Llm` endpoint/model for every child process.
3. **Starts SQL Server and Elasticsearch** (Docker) and waits for health.
4. **Builds and starts** `demo-api` , hammers its endpoints to **generate realistic error traffic**, then reshapes the logs into Elasticsearch via the ingest pipeline.
5. **Runs the CLI triage** on Qwen, then **builds and starts the API (`:5283`) and the frontend (`:5173`)**.
6. **Creates a triage run via the API** and **asserts** the result's `analyzedBy` is `Llm` / `Mixed` — i.e. it **fails loudly if Qwen was not actually used**.
7. **Opens the browser** on the produced run, where every analysed group shows the 🤖 **Qwen** badge.

8.4 Expected console output (key markers)

```
[demo] LLM strategy enabled: provider=Qwen model=qwen2.5:3b endpoint=http://localhost:11434
[qwen-bootstrap] chat probe: ok
[demo] Producing error traffic
[es-bootstrap] Reindex: total=94 created=94 updated=0 failures=0
[demo] Running CLI: hot-fix-ambulance demo-api --lookback 1h
[demo] Starting HotFixAmbulance.Api on http://localhost:5283
[demo] Creating the run via API: POST /api/triage/demo-api?lookbackHours=1
[demo] API run id=<guid> analyzedBy=Llm groups=38
[demo] Verified: this run was analyzed by Qwen (analyzedBy=Llm). The UI shows the Qwen badge.
[demo] Opening UI: http://localhost:5173/?analysisId=<guid>&api=demo-api
```

8.5 What you should see

The browser opens the triage table for `demo-api`. **Each group's "Suggestion for Error" and "How to fix" cells carry a violet 🤖 Qwen badge **with model-generated text — exactly as in** Figure 1, Figure 2 and Figure 3** above.

8.6 Verify the LLM directly (optional, independent proof)

```
# The Qwen model is loaded in the container
docker exec hfa-qwen ollama list                      # → qwen2.5:3b  ...  1.9 GB

# Produce a run via the API and confirm it was analysed by the LLM
curl.exe -s -X POST "http://localhost:5283/api/triage/demo-api?lookbackHours=1"
#  → JSON whose "analyzedBy" field is "Llm"

# Watch the Qwen container saturate the CPU during inference
docker stats hfa-qwen --no-stream                      # → CPU ~1000%+ while a run is in flight
```

8.7 Troubleshooting

- `x509: certificate signed by unknown authority` **on model pull (corporate networks).** `demo.ps1` already handles this — it exports the host CA bundle (`infra/qwen/export-host-ca.ps1`) and starts the runtime with `infra/qwen/docker-compose.corp.yml` so the container trusts an intercepting proxy. On a normal home network no action is needed.
- `docker compose up failed` **.** Docker Desktop is not running — start it and re-run.
- **A port is busy.** Stop whatever holds `5283` / `5173` / `11434` , or close a previous demo run.
- **Model pull is slow.** It is a one-time ~2 GB download; the demo waits for it. Re-runs skip it.
- **Run without the LLM** (heuristic only, no Docker model): `powershell -File scripts/demo.ps1 -WithElastic -KeepRunning -SkipLlm` .

8.8 Teardown

```
docker compose -f infra/qwen/docker-compose.yml down            # keeps the model volume
docker compose -f infra/elasticsearch/docker-compose.yml down
docker compose -f infra/mssql/docker-compose.yml down
# add `-v` to any of the above to also delete its data/model volume
```